PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Efficacy of dezocine on preventing opioids-induced cough during general anesthesia induction: a PRISMA-compliant systematic review and meta-analysis
AUTHORS	He, Li-xian; Yao, Yun-tai; Shao, Ken; Zhao, Yuan-yuan; Ma, Jie

VERSION 1 – REVIEW

REVIEWER	Sohan Solanki Tata Memorial Centre, Anaesthesiology, Critical Care and Pain
REVIEW RETURNED	01-Jun-2021

GENERAL COMMENTS	Li-xian He et al did a systematic review and meta-analysis of the effect of Dezocaine for preventing opioid-induced cough. The design and the methodology and results of this systematic review and meta-analysis is good. My points are: The completion date of this review as per the PROSPERO registry is 30th Sept 2019. In the abstract, they mentioned that the updated search was included till 31st December 2020 but in the main text, in the data strategy section they have mentioned that the updated search was included till 25th August 2020. Please explain this mismatch of data search strategies. A librarian should preferably be included in the search strategies for a thorough and complete search of databases. Although they have mentioned in limitation that all included studies were from China, they should clearly mention in limitation that results of this meta-analysis cannot be generalised to the whole world.

REVIEWER	Nathan Pace University of Utah
REVIEW RETURNED	20-Jun-2021

GENERAL COMMENTS	1. There is a nomenclature concern. "Incidence" is the number of patients with an event divided by the number of patients divided by a standardized period of time (typically one year) or (# patients with events/# patients/year). In the anesthesia literature "incidence" is commonly, but incorrectly used to describe risk. The authors should consider changing the nomenclature. 2. Page 10, LInes 7 to 9: "There were no patient withdrawal or dropout, neither selectiveness nor bias in all 33 RCTs." The authors used the Cochrane Risk of Bias tool with the appraisal of study quality displayed in Supplementary Figures 2 and 3. There is serious concerns about study quality displayed in those figures, particularly the concealment of random allocation. I encourage the authors to

	review the Cochrane Handbook on methods of assessing study quality. They should reconsider their conclusions. 3. The authors have add important text about limitations in the discussion. There is a more formal template for doing so. I encourage the authors to adopt the GRADE scoring process (multiple methods papers in J Clin Epidemiol). The quality of evidence is assigned a score High Quality, Moderate Quality, Low Quality and Very Low Quality based on five domains (study limitations (risk of bias), imprecision, inconsistency, indirectness, publication bias). The following link is very useful introductory webpage by the GRADE Working (https://gdt.gradepro.org/app/handbook/handbook.html). As an example, 25% of studies had high risk of bias for random allocation. This meta analysis might be downrated for study limitations. Please consider adding GRADE methods.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Dr. Sohan Solanki, Tata Memorial Centre

Comments to the Author:

Li-xian He et al did a systematic review and meta-analysis of the effect of Dezocaine for preventing opioid-induced cough.

The design and the methodology and results of this systematic review and meta-analysis is good.

My points are:

The completion date of this review as per the PROSPERO registry is 30th Sept 2019.

In the abstract, they mentioned that the updated search was included till 31st December 2020 but in the main text, in the data strategy section they have mentioned that the updated search was included till 25th August 2020. Please explain this mismatch of data search strategies.

Response:

Thanks for your detailed review.

The protocol of present meta-analysis was submitted to PROSPERO on 4th Jul. 2019 and was anticipated to be completed 30th Sept. 2019 registered. However, it was not registered until 11th Nov. 2019.

Actually, we completed this meta-analysis on 31th December 2019. From the date of completion to this submission, we went through submitting, rejecting, updating, resubmitting over and over again.

Because of the last submission to another journal on 30th August 2020, we once updated our search strategy on 25th August 2020. Before this submission to BMJ Open, we updated our search strategy on 31th December 2020 again but not in the main text. We are sorry for our carelessness and the mismatch of data search strategies has been revised.

A librarian should preferably be included in the search strategies for a thorough and complete search of databases.

Response:

Thanks for your suggestion. To be honestly, the librarian for medical research is not available in our hospital. Furthermore, our team has been systematically and professionally trained in the field of meta-analysis, including comprehensive literature search. Our published meta-analyses are as follows.

[1] Yao YT, **He LX**, Fang NX, Ma J. Anesthetic Induction With Etomidate in Cardiac Surgical Patients: A PRISMA-Compliant Systematic Review and Meta-Analysis. **J Cardiothorac Vasc Anesth**. 2021 Apr;35(4):1073-1085.

[2] Yao YT, **He LX**, Zhao YY. The effect of levosimendan on postoperative bleeding and blood transfusion in cardiac surgical patients: a PRISMA-compliant systematic review and meta-analysis. **Perfusion**. 2021 Oct;36(7):694-703.

[3] Yao YT, **He LX**, Tan JC. The effect of tranexamic acid on the values of activated clotting time in patients undergoing cardiac surgery: A PRISMA-compliant systematic review and meta-analysis. **J Clin Anesth**. 2020 Dec;67:110020.

Although they have mentioned in limitation that all included studies were from China, they should clearly mention in limitation that results of this meta-analysis cannot be generalised to the whole world.

Response:

Actually, we have mentioned in Discussion and limitation part as: Secondly, all the 33 included RCTs were performed in China. The anti-tussive effectiveness of DZC in other ethnicities remains to be investigated.

As you suggested, we have rewritten the sentence as: Secondly, all the 33 included RCTs were performed in China. The anti-tussive effectiveness of DZC may not be generalised to the whole world and remains to be investigated in other ethnicities.

Reviewer: 2

Dr. Nathan Pace, University of Utah

Comments to the Author:

1. There is a nomenclature concern. "Incidence" is the number of patients with an event divided by the number of patients divided by a standardized period of time (typically one year) or (# patients with events/# patients/year). In the anesthesia literature "incidence" is commonly, but incorrectly used to describe risk. The authors should consider changing the nomenclature.

Response:

Thank you for your suggestion. We have replaced **incidence** with **occurrence rate**.

2. Page 10, LInes 7 to 9: "There were no patient withdrawal or dropout, neither selectiveness nor bias in all 33 RCTs." The authors used the Cochrane Risk of Bias tool with the appraisal of study quality displayed in Supplementary Figures 2 and 3. There is serious concerns about study quality displayed in those figures, particularly the concealment of random allocation. I encourage the authors to review the Cochrane Handbook on methods of assessing study quality. They should reconsider their conclusions.

Response:

We have tone down the Conclusion as editor's suggestion and rewritten the text as DZC significantly inhibited OIC and may be used to manage OIC induced by fentanyl or sufentanil. More high-quality RCTs are needed to complement the safety of DZC.

3. The authors have add important text about limitations in the discussion. There is a more formal template for doing so. I encourage the authors to adopt the GRADE scoring process (multiple methods papers in J Clin Epidemiol). The quality of evidence is assigned a score High Quality, Moderate Quality, Low Quality and Very Low Quality based on five domains (stud y limitations (risk of bias), imprecision, inconsistency, indirectness, publication bias). The following link is very useful introductory webpage by the GRADE Working (https://gdt.gradepro.org/app/handbook/handbook.html). As an example, 25% of studies had high risk of bias for random allocation. This meta analysis might be downrated for study limitations. Please consider adding GRADE methods.

Response:

As you suggested, GRADE scoring has been presented in **Table 2** and **3**, for assessment for primary and secondary outcomes respectively.